

GRANT COUNTY AP, NM, USA

WMO: 722721

Lat: 32.633N

Lon: 108.167W

Elev: 1638

StdP: 83.14

Time Zone: -7.00 (NAM)

Period: 96-19

WBAN: 93063

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -7.5	(c) -5.2	(d) -20.3	(e) 0.7	(f) 6.4	(g) -18.0	(h) 0.9	(i) 9.0	(j) 12.7	(k) 6.7	(l) 11.6	(m) 8.2	(n) 2.9	(o) 340	(p) 0.529

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 17.5	(c) 36.9	(d) 16.1	(e) 34.9	(f) 15.6	(g) 33.1	(h) 15.4	(i) 20.2	(j) 27.1	(k) 19.6	(l) 26.3	(m) 18.9	(n) 25.8	(o) 4.8	(p) 310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.6	16.5	22.3	17.7	15.5	21.6	17.2	15.0	21.6	65.7	27.3	63.3	26.2	60.9	25.4	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.7	10.2	8.7	-11.5	38.2	3.3	1.8	-13.9	39.5	-15.8	40.6	-17.6	41.6	-20.0	43.0		
			-12.9	20.7	3.2	1.4	-15.2	21.7	-17.0	22.5	-18.8	23.3	-21.1	24.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.4	5.7	7.6	11.2	14.6	18.8	24.6	25.5	24.0	21.1	15.6	9.9	5.4
	DBStd	7.93	3.79	4.30	3.56	3.54	3.91	2.70	2.49	2.12	2.53	3.92	4.12	4.02
	HDD10.0	468	139	86	28	7	1	0	0	0	6	52	150	
	HDD18.3	1838	393	301	221	118	43	0	0	0	6	101	253	402
	CDD10.0	2424	5	19	66	144	272	439	479	434	332	179	49	6
	CDD18.3	751	0	0	0	5	57	189	221	176	88	17	0	0
	CDH23.3	9783	0	11	60	296	1151	2787	2368	1727	1009	320	53	1
	CDH26.7	4503	0	1	7	63	478	1534	1204	760	367	84	5	0
(14) Wind	WSAvg	4.0	3.7	4.2	4.5	4.9	4.5	4.4	3.9	3.4	3.4	3.6	3.6	3.6
(15) Precipitation	PrecAvg	405	31	27	16	11	13	19	80	78	50	28	20	31
	PrecMax	524	125	105	49	69	92	87	149	148	82	101	100	110
	PrecMin	245	5	0	2	2	0	0	26	46	2	0	0	0
	PrecStd	71	30	23	12	14	21	18	35	29	24	24	23	29
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.9	24.8	27.5	31.0	36.1	38.2	38.0	36.0	33.8	32.1	27.4	22.0
		MCWB	8.6	9.5	9.9	11.1	14.3	15.7	16.6	17.5	15.6	15.2	11.5	9.6
	2%	DB	18.7	22.1	25.0	28.0	32.9	37.1	36.3	33.8	32.3	28.9	24.9	18.7
		MCWB	7.2	8.4	9.1	10.4	13.2	15.4	17.4	16.9	15.8	14.6	10.4	7.4
	5%	DB	16.8	19.1	22.9	27.0	31.4	35.2	34.1	32.5	30.6	27.2	22.4	17.0
		MCWB	6.3	7.2	8.4	10.0	12.4	14.6	16.8	17.1	15.7	13.2	9.6	6.6
	10%	DB	14.1	17.4	21.3	24.9	29.1	33.7	32.7	31.1	28.7	24.8	19.7	13.9
		MCWB	5.2	6.4	7.7	9.2	11.6	14.3	16.7	17.2	15.2	11.8	8.7	5.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.9	10.8	11.3	13.0	15.7	18.9	21.4	21.1	19.7	18.0	12.7	11.2
		MCDB	13.7	19.8	22.5	24.5	28.6	27.1	28.2	27.9	26.7	25.9	22.5	17.9
	2%	WB	9.1	9.8	10.0	11.6	14.7	17.8	20.5	20.2	18.6	16.4	11.4	9.0
		MCDB	15.0	18.1	21.8	24.8	28.2	28.8	27.5	26.9	24.9	24.2	22.2	15.8
	5%	WB	7.6	8.5	9.1	10.6	13.5	16.7	19.8	19.6	17.9	15.1	10.4	7.5
		MCDB	13.9	15.9	20.7	24.5	28.0	29.5	26.5	26.1	24.2	22.7	20.5	15.2
	10%	WB	6.3	7.4	8.3	9.8	12.4	15.9	19.1	18.9	17.2	13.9	9.3	6.2
		MCDB	13.1	15.4	19.6	23.4	27.6	29.5	26.2	25.5	23.7	22.0	18.3	13.1
(35) Mean Daily Temperature Range	5% DB	MDBR	15.6	16.2	17.3	18.0	18.3	17.5	13.8	13.5	14.4	16.5	16.8	15.2
		MCDBR	19.4	20.1	20.1	20.6	20.3	18.5	16.0	15.6	17.1	18.4	20.9	18.8
	5% WB	MCWBR	10.7	9.9	9.4	9.5	8.6	7.2	4.6	4.4	5.8	8.0	10.4	10.3
		MCWBR	16.0	16.2	18.0	18.7	18.6	16.2	13.1	12.4	12.6	15.0	18.6	16.2
(40) Clear-Sky Solar Irradiance	taub		0.237	0.241	0.259	0.280	0.296	0.322	0.366	0.344	0.311	0.269	0.246	0.237
		taud	2.693	2.694	2.598	2.513	2.515	2.439	2.424	2.457	2.549	2.649	2.678	2.686
	Ebn at Noon		1016	1038	1030	1010	988	957	913	932	958	988	995	998
		Edh at Noon	70	77	92	105	106	115	116	110	96	80	70	67
(44) All-Sky Solar Radiation	RadAvg		3.54	4.55	5.99	7.33	8.06	8.00	6.93	6.51	5.86	4.94	3.90	3.25
	RadStd		0.29	0.40	0.35	0.29	0.32	0.45	0.33	0.20	0.29	0.38	0.21	0.16

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (10 neighbors)	+0.30	N/A	N/A	N/A	N/A	N/A	N/A	-75	+90	N/A

Nomenclature: See separate page